

RBG-48261

10 CFR 50.73

November 15, 2023

ATTN: Document Control Desk
U.S. Nuclear Regulatory Commission
Washington, DC 20555-0001

Subject: Licensee Event Report 50-458 / 2022-004-01, High Pressure Core Spray
Inoperable due to Transformer Failure

River Bend Station – Unit 1
NRC Docket Nos. 50-458
Renewed Facility Operating License No. NPF-47

On November 17, 2022, Licensee Event Report 50-458 / 2022-004-00, High Pressure Core Spray Inoperable due to Transformer Failure (ML22321A306) was submitted by Entergy Nuclear Operating Inc. This letter is a supplement to the report to provide the updated results of the causal evaluation. The updated information is denoted by revision bars located in the right-hand margin. The report is submitted in accordance with 10 CFR 50.73.

This document contains no commitments.

Should you have any questions, please contact Mr. Randy Crawford, Regulatory Assurance Manager, at 225-381-4177.

Respectfully,



Randy Crawford

RC/twf

Enclosure: Licensee Event Report 50-458 / 2022-004-01, High Pressure Core Spray
Inoperable due to Transformer Failure

cc: NRC Region IV Regional Administrator - Region IV
NRC Senior Resident Inspector - River Bend Station

Enclosure

RBG-48261

**Licensee Event Report 50-458 / 2022-004-01, High Pressure Core Spray Inoperable due to
Transformer Failure**

**LICENSEE EVENT REPORT (LER)**

(See Page 2 for required number of digits/characters for each block)

(See NUREG-1022, R.3 for instruction and guidance for completing this form
<http://www.nrc.gov/reading-rm/doc-collections/nuregs/staff/sr1022/r3/>)

Estimated burden per response to comply with this mandatory collection request: 80 hours. Reported lessons learned are incorporated into the licensing process and fed back to industry. Send comments regarding burden estimate to the FOIA, Library, and Information Collections Branch (T-6 A10M), U. S. Nuclear Regulatory Commission, Washington, DC 20555-0001, or by email to Infocollections.Resource@nrc.gov, and the OMB reviewer at: OMB Office of Information and Regulatory Affairs, (3150-0104), Attn: Desk Officer for the Nuclear Regulatory Commission, 725 17th Street NW, Washington, DC 20503; email: opa_submission@omb.eop.gov. The NRC may not conduct or sponsor, and a person is not required to respond to, a collection of information unless the document requesting or requiring the collection displays a currently valid OMB control number.

1. Facility Name

River Bend Station, Unit 1

**050****052****2. Docket Number**

458

3. Page

1 OF 3

4. Title

High Pressure Core Spray Inoperable due to Transformer Failure

5. Event Date			6. LER Number			7. Report Date			8. Other Facilities Involved		
Month	Day	Year	Year	Sequential Number	Revision No.	Month	Day	Year	Facility Name	☐ 050	Docket Number
09	19	2022	2022	- 004 -	01	11	15	2023	N/A	☐ 050	N/A
									N/A	☐ 052	N/A

9. Operating Mode

1

10. Power Level

100%

11. This Report is Submitted Pursuant to the Requirements of 10 CFR §: (Check all that apply)

10 CFR Part 20	☐ 20.2203(a)(2)(vi)	10 CFR Part 50	☐ 50.73(a)(2)(ii)(A)	☐ 50.73(a)(2)(viii)(A)	☐ 73.1200(a)
☐ 20.2201(b)	☐ 20.2203(a)(3)(i)	☐ 50.36(c)(1)(i)(A)	☐ 50.73(a)(2)(ii)(B)	☐ 50.73(a)(2)(viii)(B)	☐ 73.1200(b)
☐ 20.2201(d)	☐ 20.2203(a)(3)(ii)	☐ 50.36(c)(1)(ii)(A)	☐ 50.73(a)(2)(iii)	☐ 50.73(a)(2)(ix)(A)	☐ 73.1200(c)
☐ 20.2203(a)(1)	☐ 20.2203(a)(4)	☐ 50.36(c)(2)	☐ 50.73(a)(2)(iv)(A)	☐ 50.73(a)(2)(x)	☐ 73.1200(d)
☐ 20.2203(a)(2)(i)	10 CFR Part 21	☐ 50.46(a)(3)(ii)	☐ 50.73(a)(2)(v)(A)	10 CFR Part 73	☐ 73.1200(e)
☐ 20.2203(a)(2)(ii)	☐ 21.2(c)	☐ 50.69(g)	☐ 50.73(a)(2)(v)(B)	☐ 73.77(a)(1)	☐ 73.1200(f)
☐ 20.2203(a)(2)(iii)		☐ 50.73(a)(2)(i)(A)	☐ 50.73(a)(2)(v)(C)	☐ 73.77(a)(2)(i)	☐ 73.1200(g)
☐ 20.2203(a)(2)(iv)		☐ 50.73(a)(2)(i)(B)	☒ 50.73(a)(2)(v)(D)	☐ 73.77(a)(2)(ii)	☐ 73.1200(h)
☐ 20.2203(a)(2)(v)		☐ 50.73(a)(2)(i)(C)	☐ 50.73(a)(2)(vii)		

☐ OTHER (Specify here, in abstract, or NRC 366A).**12. Licensee Contact for this LER****Licensee Contact**

Randy Crawford, Manager – Regulatory Assurance

Phone Number (Include area code)

225-381-4177

13. Complete One Line for each Component Failure Described in this Report

Cause	System	Component	Manufacturer	Reportable to IRIS	Cause	System	Component	Manufacturer	Reportable to IRIS
B	EB	XFMR	G080	Y	N/A	N/A	N/A	N/A	N/A

14. Supplemental Report Expected☒ No ☐ Yes (If yes, complete 15. Expected Submission Date)**15. Expected Submission Date**

Month	Day	Year
N/A	N/A	N/A

16. Abstract (Limit to 1326 spaces, i.e., approximately 13 single-spaced type written lines)

At 0132 CDT on September 19, 2022, River Bend Station (RBS) was operating at 100% power when the High Pressure Core Spray System (HPCS) was declared inoperable due to a transformer fault that occurred during Division III Standby Diesel Generator monthly surveillance testing. RBS entered Technical Specification 3.8.9, Condition E and declared HPCS and Standby Service Water System Pump 2C (SWP-P2C) inoperable. Reactor Core Isolation Cooling (RCIC) was verified operable. HPCS is a single train system at RBS, therefore an 8-hour report was submitted in accordance with 10 CFR 50.72(b)(3)(v)(D) as an event or condition that could have prevented the fulfillment of a safety function (EN 56116). This event is also being reported in accordance with 10 CFR 50.73(a)(2)(v)(D) as an event or condition that could have prevented the fulfillment of a safety function.

The condition was corrected by replacing the transformer on September 30, 2022. HPCS and SWP-P2C were restored to operable at 0921 CDT on September 30, 2022.

**LICENSEE EVENT REPORT (LER)
CONTINUATION SHEET**

(See NUREG-1022, R.3 for instruction and guidance for completing this form
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1. FACILITY NAME River Bend Station, Unit 1	☒ 050	2. DOCKET NUMBER 458	3. LER NUMBER		
	☐ 052		YEAR 2022	SEQUENTIAL NUMBER - 004	REV NO. - 01

NARRATIVE**EVENT DESCRIPTION**

At 0132 CDT on September 19, 2022, River Bend Station (RBS) was operating at 100% power. There were no other inoperable structures, systems, or components (SSCs) that contributed to this event. The Division III Diesel Generator [DG] was started per STP-309-0203, DIVISION III DIESEL GENERATOR OPERABILITY TEST. Approximately 5 seconds later the Main Control Room received multiple alarms on Division 3 components. The field operator performed a visual inspection of the Division III switchgear and discovered that the E22-S003, HPCS TRANSFORMER [XFMR] was damaged. The Division III Diesel Generator was secured and placed in maintenance mode.

The transformer failure caused a loss of power to E22-S002, Division III 480 VAC switchgear [SWGR]. E22-S002, Division III 480 VAC switchgear supplies power to HPCS motor operated valves that are required to align flow to the reactor pressure vessel following system initiation. RBS entered Technical Specification (TS) 3.8.9, Condition E and immediately declared the High Pressure Core Spray System (HPCS) [BG] and Standby Service Water System Pump 2C (SWP-P2C) [P] inoperable. This also placed RBS in a 14-day TS LCO shutdown action statement due to the inoperability of HPCS. Reactor Core Isolation Cooling (RCIC) [BN] was verified operable.

On June 21, 2022, Preventive Work Order 53003640 was performed on the transformer. No signs of degradation were noted, and all acceptance criteria were met satisfactory. Following the completion of the preventive work and prior to the transformer fault, STP-309-0203, DIVISION III DIESEL GENERATOR OPERABILITY TEST was successfully completed on July 25, 2022, and again on August 22, 2022.

Prior to this event HPCS and the Division 3 Transformer power source were operable. At time of discovery, the HPCS lost its initiation capability because the E22-S002, Division III 480 VAC switchgear would not have been powered from the Division III Standby Diesel Generator on any HPCS initiation signal. HPCS was inoperable and incapable of performing its design function during this event. RBS is reporting the event in accordance with 10 CFR 50.73(a)(2)(v)(D) as an event or condition that could have prevented the fulfillment of a safety function.

SAFETY ASSESMENT

The RBS UFSAR states that in the event of a HPCS system failure such that HPCS cannot maintain reactor water level, the Automatic Depressurization System (ADS), which is independent of any other Emergency Core Cooling System (ECCS), reduces the reactor pressure so that flow from Low Pressure Coolant Injection (LPCI) [BO] and Low Pressure Core Spray (LPCS) [BM] systems enters the reactor vessel in time to cool the core and limit fuel cladding temperature. Additionally, high pressure makeup was available from RCIC. Two of three divisions of ECCS are required for the RBS loss of coolant accident analysis. While HPCS was out of service, Division 1 ECCS and Division 2 ECCS systems and ADS were operable and would have met the ECCS performance criteria of 10CFR50.46. The out of service time was maintained within the limits of RBS Technical Specifications.

There were no actual Nuclear or Radiological safety consequences due to this event. This event was of minimal significance to the health and safety of the public.

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NARRATIVE**EVENT CAUSE**

A Root Cause Evaluation (RCE) and an Equipment Failure Evaluation determined that the direct cause for the E22-S003 HPCS Transformer failure was an open circuit fault initiated by separation of the threaded stud at the secondary bus brazed connection, requiring replacement of the transformer.

The RCE determined the root cause for the HPCS Transformer failure was a cable to bus bar connection design that was susceptible to degradation from routine maintenance.

The HPCS Transformer Low Side (secondary) connections to the bus bars were made by brazing threaded brass studs to the bus bars. The bus bars were located behind insulated material. The threaded brass studs protruded through holes in the insulation material and each phase was connected by a bolted arrangement.

The brazed connections to the bus bar were degraded over time during routine maintenance testing activities that disassembled the bolted connections. The configuration did not provide a provision for installing a backing wrench to prevent the threaded studs from rotating when the connection nuts are being loosened or tightened. Torsional forces were being applied to the brazed stud connections each time the connection was disassembled and reassembled.

CORRECTIVE ACTIONS

Complete:

Replacement transformer was installed with a Temporary Modification Engineering Change. The bus connection for the temporary replacement transformer is bolted.

Planned actions tracked in the Corrective Action Program:

- Complete the procurement and installation of the permanent replacement HPCS transformer.
- Update the Design Specification to include transformer's termination of connectors for leads on low and high voltage sides should be bolted and accessible for maintenance.

PREVIOUS SIMILAR EVENTS

None

NOTE:

Energy Industry Identification System and Component Function Identification codes are identified in the text of this report following the first mention of the system or component as [XX].